

Motor Imagery EEG Recognition Based on Weight-Sharing CNN-LSTM Network

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Abstract: The brain-computer interface technology enables the disabled to control external devices through the motor imagery EEG. Due to the complex changes of EEG in the time domain and frequency domain, classifiers play an important role in EEG recognition. Convolutional neural network is an excellent deep learning method, but most papers usually use one-dimensional convolution to identify EEG, and rarely consider comprehensive feature extraction and classification of time-frequency map through two-dimensional convolutional network. In this article, the time-frequency graphs of different EEG channels are superimposed by referring to the color dimension of the picture. A weighted shared two-dimensional convolutional CNN-LSTM network is proposed, which shares convolution kernels for feature maps of different channels. Compared with CNN and CNN-LSTM, the weight-sharing CNN-LSTM reduces the amount of calculation, speeds up the network training and improves the classification performance, the highest accuracy rate is 82.3%.

Key Words: EEG, BCI, Weight-Sharing CNN-LSTM, Random Superposition, Wavelet Transform, Classification

1 INTRODUCTION

As a physiological signal that reflects the activity of the brain, electroencephalogram (EEG) plays an important role in epilepsy detection [1, 2], depression [3], motor imagery [4, 5], sentiment analysis [6, 7], person-discriminative biometric [8] and sleep detection [9]. Motor imagery brain-computer interface (BCI) realizes non-invasive control of external devices, such as mechanical arm and keyboard, by means of motor EEG, providing a safe and reliable helps for the disable. The general EEG analysis process include preprocessing, feature extraction and classification. In addition to the traditional feature power spectral density (PSD)[10], the input features of the deep learning network also include the angle-amplitude transformation (AAT) [11] and two-dimensional gray-scale image conversion based on isometric azimuth projection [4] which are newly proposed [6] can get better result by using multi-channel raw signals [12, 13] or 3D-grid features superimposed after multiple segments of time series as input. Research in recent years has shown that Hybrid Architecture Network is better than the single network. Tabar Y R [14] et al. investigated CNN and stacked autoencoders (SAE) to classify EEG Motor Imagery signals.[15]proposed a multi-layer CNNs method for fusing CNNs with different characteristics and architectures. Tan, Chuanqi et al. [4] com-

bined CNN and RNN for feature extraction and classification. Khosropanah et al. achieved pre-surgical evaluation of epilepsy through Adaptive neuro-fuzzy inference system[16]. [8] proposed an adversarial reasoning method for biometric recognition.

When humans perform motor imagination, the EEG rhythm energy in the contralateral motor sensory area of the cerebral cortex will be significantly reduced, while the EEG rhythm energy in the ipsilateral motor sensory area increases, which is called Event Related Desynchronization/Event Related Synchronization (ERD/ERS). According to the definition of ERD/ERS, the spatial domain features of EEG are very important for the recognition of motor imagination, but due to the complexity of EEG, most studies can only consider the time-frequency domain or the time domain and the spatial domain [5] etc. For example, the time-frequency features of different channels are pieced together in the frequency domain to obtain a feature map as the input of a one-dimensional CNN [1] or the original signal as the input without involving frequency domain features. This paper proposes a new combination method, which regards the time-frequency graphs of different EEG channels as color channels for image processing. The four-dimensional network inputs are samples, frequency features, time domain features and spatial domain feature. Therefore, this paper designs a CNN-LSTM network that shares the weights in the spatial, time and frequency domains to classify EEG signal. The process of EEG analysis in this article is data preprocessing, feature extraction, network introduction, experimental results and

This work is supported in part by the National Key Research and Development Program of China (grant no. 2017YFB1401203), National Natural Science Foundation (NNSF) of China under Grant 62173032, 62003038, and Foshan Science and Technology Innovation Program (grant no. BK20AF005).

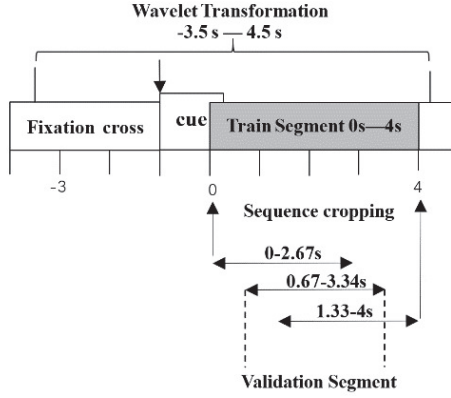


Figure 1: Training set division and Sequence Cropping

analysis, conclusions. Section I is mainly a brief introduction and literature review of EEG signal analysis. The signal preprocessing is described in Section II. Section III introduces the feature extraction and results based on wavelet transform. Section IV describes the proposed Weight Sharing CNN-LSTM network in this paper. Section V lists the classification results and network parameters for comparison. Section VI analyzes the classification results and summarizes the full text.

2 PREPROCESSING

First, this article performs 6~25Hz band-pass filtering on the EEG. Then the sampling frequency is down-sampled from 160Hz to 60Hz. -3.5s to 4.5s EEG signal is cropped and wavelet transformed, finally time-frequency graphs are obtained. Among them, the 0~4s of the training sample will be trained and the test sample only takes 0.67s~3.34s, which is the middle part of the third halves of 0~4s, as shown in Fig.1. Because the number of samples is too small, this article uses three data augmentation methods, namely Segmentation and Recombination, Sequence Cropping and Random Superposition.

2.1 Sequence Cropping

As shown in Fig.1, the time-frequency graph of size 17×120 is cropped sequentially, with a window size of 17×80 and a step size of 20. The training data set is doubled by $(120 - 80) / 20$ and the size of the time-frequency graph becomes 17×80 .

2.2 Segmentation and Recombination

Segmentation and Recombination refers to randomly selecting a group of EEG signals $(S_1, S_2, \dots, S_N)$, where S_i represents a segment of EEG of length M , $i = 1, 2, 3, \dots, N$. Dividing them into four equal parts and obtaining four groups of EEG signals of length $M/4$, $(S_1^1, S_2^1, \dots, S_N^1)$, $(S_1^2, S_2^2, \dots, S_N^2)$, $(S_1^3, S_2^3, \dots, S_N^3)$ and $(S_1^4, S_2^4, \dots, S_N^4)$. Then randomly scramble the four groups of EEG signals and combine them to obtain new EEG signal samples $(S'_1, S'_2, \dots, S'_N)$.

2.3 Random Superposition

Random Superposition refers to randomly selecting N groups of training samples, where the quantity of samples and sample numbers contained in groups are randomly selected. Then the EEG signals of each group are added and averaged to obtain a new training sample. This article randomly superimposes the EEG and the time-frequency graphs to increase the number of samples.

For the Random Superposition of EEG, since the motor imagery EEG is time-locked and non-phase-locked induction reactions, when the superimposed sample is too large, the induction reactions in different EEG will cancel each other out. Therefore, the number of samples in each group is limited to 5~20. The number of samples to be superimposed is randomly selected. Then the EEG signal samples are randomly selected and a new EEG signal sample is obtained after averaging them. The Random Superposition of the time-frequency graphs is similar with the above operation. The superposition number is randomly selected. Some features are randomly selected and averaged to obtain a new time-frequency map. The horizontal axis of the time-frequency graph represents time, the vertical axis represents frequency and the value represents the energy level, which reflects the signal energy level of the EEG signal in a certain frequency band at a certain time. Random Superposition based on time-frequency graphs does not need to be limited in number, because the time-frequency graph reflects the energy change of the EEG signal relative to the baseline, which will not disappear with the superposition but increase.

3 Wavelet Transform

The time-frequency graph reflects the energy change of the signal at different frequencies as time increases. Wavelet Transform[17, 16] is a time-frequency localization analysis method in which both the time window and the frequency window can be changed. It can characterize the local characteristics of the signal in the time domain and the frequency domain. For the signal sequence $X = \{x(0), x(1), \dots, x(N-1)\}$, the continuous wavelet transform can be expressed as

$$\int_{-\infty}^{+\infty} x(t) \frac{1}{\sqrt{a}} \psi \times \left(\frac{t-b}{a} \right) dt \quad (1)$$

In the formula 1, $\Psi(t)$ is the wavelet function; the parameter a is the scale factor, which stretches the wavelet function; the parameter b represents the displacement, usually $a = 1 \setminus 2j$, $b = k \setminus 2j$.

In this paper, Morlet Wavelets are used as the basis function to perform wavelet transformation on the 0~4s of the signal, the frequency domain features of 8~25Hz are cropped and a 17×120 time-frequency diagram is obtained. As shown in Fig.2, the first three represent the time-frequency diagrams of the left-hand motor imagery and the last three represent the time-frequency diagrams of the right-hand motor imagery.

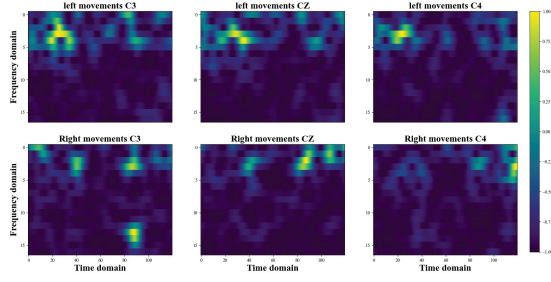


Figure 2: Time-frequency diagram of C3, CZ, C4 channels of EEG of left-right hand movement imagery

4 Deep Learning Network

4.1 Convolutional Neural Networks

Convolutional Neural Network is a multi-layer neural network structure with convolutional layer, pooling layers and fully connected layers. The training process can be divided into the following steps:

(a) The two-dimensional input passes through the convolutional layer and Rectified Linear Unit (ReLU) layer of the neural network.

$$h_{ij}^k = f((W^k * x)_{ij} + b^k) \quad (2)$$

x is the input; W^k is the weight; b^k is the bias. The gradient calculation of convolutional neural networks is:

$$W^k = W^k - \eta \frac{\partial E}{\partial W^k} \quad (3)$$

$$b_k = b_k - \eta \frac{\partial E}{\partial b_k} \quad (4)$$

η represents the learning rate.

4.2 Long Short Term

Long Short Term (LSTM) is a special type of RNN, which is often used for long sequence classification. It consists of Forget gates, Input gates and Output gates.[18]

The Forget Gate uses decision vector f_t to determine how much of the cell state C_{t-1} at the previous moment is retained to the current moment cell state C_t , 0 means discard and 1 means pass.

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \quad (5)$$

σ is the sigmoid activation function; W_f is the weight of the Forget gate of dimension $d_c \times (d_h + d_x)$; $[h_{t-1}, x_t]$ represents the output value h at $t-1$ and the input x at time t splicing; b_f represents the forget gate offset.

The Input Gate determines how much of the current network input x_t is stored in the unit state C_t by multiplying decision vector i_t with the candidate information $\bar{C}_t$.

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i) \quad (6)$$

$$\bar{C}_t = \tanh(W_c[h_{t-1}, x_t] + b_c) \quad (7)$$

The new cell state C_t is the combination of C_{t-1} and $\bar{C}_t$.

$$C_t = C_{t-1} * f_t + \bar{C}_t * i_t \quad (8)$$

The Output Gate controls the decision vector O_t to determine how much of the information vector C_t is output to the current output value h_t of the LSTM. Get the output O_t

$$O_t = \sigma(W_o[h_{t-1}, x_t] + b_o) \quad (9)$$

Output values of LSTM

$$h_t = \tanh(C_t) * O_t \quad (10)$$

4.3 Weight Sharing CNN-LSTM

This article refers to the RGB channels in image processing to stack the time-frequency graphs of different EEG channels. The weight-sharing CNN-LSTM(WSCNN-LSTM) network consists of input layer, convolutional layers, LSTM units, fully connected layer and output layer. For the 17*80 feature map, it represents the channel-frequency-time feature. Unlike the traditional method of superimposing the time-frequency diagrams of different EEG channels into one picture as a network input, this article arranges the time-frequency graphs of the three EEG channels in the way of convolutional layer channels. As shown in Fig.3, the time-frequency graph of the first EEG channel is regarded as the input of the network, after convolution kernels, batch normalization layer (BN), ReLU layer and average pooling layer to get the feature map. At the same time, the other two EEG channels share these weights.

5 Experimental Results and Analysis

In this paper, BCI competition IV 2b dataset is used as the dataset to classify the motor imagery EEG based on deep learning. The dataset includes three EEG channels of C3, C4, Cz channels of 9 subjects, with a sampling frequency of 160Hz. Each subject data set has three sessions. In the first two sessions, the subject heard the buzzer and then saw the fixed cross displayed on the screen. After a few seconds, the cross on the screen pointed to the left or right for 1.25 seconds and the subject began to imagine left (right) movements. After 4 seconds or more, the subjects simply rested for 1.5 seconds or 2.5 seconds.

In order to explore the impact of data volume on model performance, this paper performed 0%, 25%, 50% and 100% data augmentation on 420 sample sets of subject5. This article mainly set 0%, 25%, 50% and 100% increase ratio, and the operation method is as described above. The experiment is shown in Table 1. The classification effect of the model with data augmentation is better than that without data enhancement, which shows that data augmentation has greatly improved the classification effect and the accuracy rate has increased by 5.95% at the highest. The average cross-validation result of the 100% increase ratio among the four increase ratios is the best, so in the end this paper selects the data increase ratio 100%.

In order to observe the convergence process of the weight-sharing CNN-LSTM network, Fig. 4 shows the curves of various indicators in the training process of the subject4 data set. The red curve represents the training curve and the blue curve represents the validation curve. It can be seen that the training loss approaches 0 at about 30 epochs

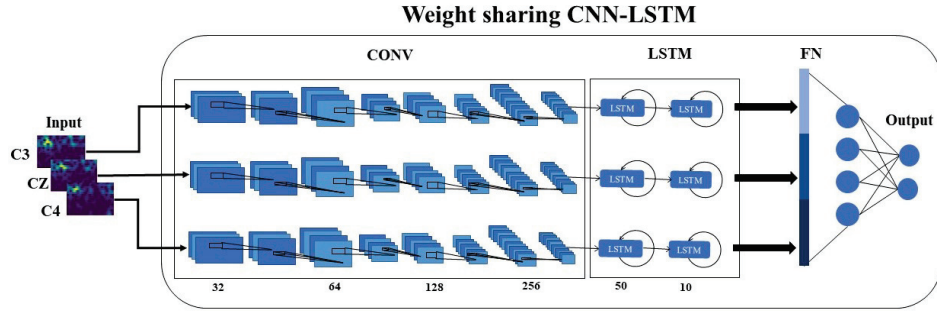


Figure 3: Weight-Sharing CNN-LSTM network structure

Table 1: Comparison of data increase ratio

No.	Appending Proportion%			
	0%	25%	50%	100%
1	80.95%	85.71%	81.75%	88.10%
2	76.19%	78.57%	73.81%	88.10%
3	75.40%	83.33%	75.40%	80.95%
4	70.64%	76.19%	73.01%	80.95%
5	74.60%	80.95%	74.60%	83.33%
6	84.13%	88.10%	85.71%	85.71%
7	78.57%	80.95%	77.78%	78.57%
8	73.81%	78.57%	76.98%	83.33%
9	80.95%	83.33%	81.75%	83.33%
10	83.33%	80.95%	81.75%	85.71%
Average	77.86%	81.67%	78.25%	83.81%

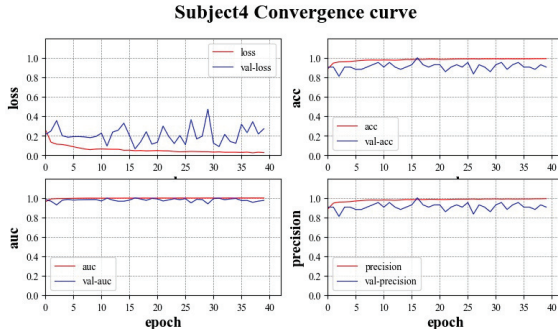


Figure 4: Subject 4 Convergence curve

and the training accuracy rate approaches 1. As the epoch increases, the val-loss gradually rises, indicating that the model has begun to overfit. For insurance, this article sets 40 epochs.

This paper compares the classification performance of the network through the ten-fold cross-validation accuracy of the nine subjects' datasets. As shown in Table 2, the classification effect of CNN is poorer than CNN-LSTM and weight-sharing CNN-LSTM, indicating that the combination of LSTM and CNN improves the classification accuracy of the network, which verifies the classification performance of hybrid network is better than the single network structure. Compared with the traditional feature stitching CNN-LSTM, the weight-sharing CNN-LSTM proposed in this paper has the highest average accuracy rate of

82.3%, which is higher than CNN and CNN-LSTM. It shows the importance of weight-sharing and feature arrangement for the classification of time-frequency graphs of EEG. This paper also compares weight-sharing CNN-LSTM with other networks using BCI competition IV 2b dataset, as shown in Table 2. Although CNN-SAE [14], mixed-scale CNN architecture [19] and RSMM[20] have a slightly higher classification effect in some certain subjects, the average 10-fold cross-validation accuracy of WS-CNN-LSTM is higher.

6 Conclusions

This article proposes a data enhancement method of Random Superposition and combines Sequence Cropping, Segmentation and Recombination to increase the training dataset. Experimental results show that data augmentation can effectively improve the accuracy of the classifier. This is because Deep Learning is data-based representation learning method. Therefore, the diversity of EEG data enhancement methods is very important. The research results also show that the classification performance of weight-sharing CNN-LSTM is better than CNN-LSTM, CNN-SAE, CNN and Mixed-scale CNN. This indicates that the combination of different structures enables the hybrid network to mine features in the time domain, frequency domain and spatial domain. Compared with WS-CNN-LSTM, CNN-LSTM also has feature sharing, but it is very likely to use the splicing of feature maps as important features, which reduces the classification performance. The dataset used in this article is the motor imagery EEG with only three channels, but there are more EEG channels in the actual application areas, such as sleep detection and emotion recognition. Hence the EEG signal characteristics of more channels should be explored.

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Table 2: Ten-fold cross-validation results of different subjects on different networks

No.	Accuracy%					
	CNN	CNN-LSTM	WS-CNN-LSTM	CNN-SAE	Mixed-scale CNN	RSMM
1	82.45±5.1	83.3±8.2	84.3±6.2	76.0±2.7	81.0±3.1	72.50
2	69.00±9.1	75.0±6.6	77.5±8.9	65.8±1.9	53.5±1.2	56.43
3	62.50±7.8	66.9±9.3	66.9±9.2	75.3±1.8	55.3±4.7	55.63
4	97.00±2.1	97.9±1.8	97.9±1.7	95.3±0.4	97.2±1.2	97.19
5	81.00±4.6	82.9±5.0	83.8±3.9	83.0±1.4	91.3±2.8	88.44
6	81.00±8.1	82.0±7.7	85.5±6.5	79.5±2.5	86.3±1.2	78.75
7	78.75±4.9	80.3±4.2	80.0±4.7	74.5±1.8	89.5±2.4	77.50
8	80.00±4.2	82.9±5.6	83.7±4.7	75.3±2.6	89.7±2.5	91.88
9	81.00±6.5	80.5±4.2	80.8±7.5	73.3±3.6	89.9±2.3	83.44
AVG	79.19±5.8	81.28±5.8	82.3±5.9	77.6±2.1	81.5±2.3	77.97

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